Electrical and Computer Engineering
University of Thessaly

ECE214 - Introduction to Electronics

Static (DC) Analysis of CMOS Inverter

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Assignment 2

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Exercise 1

1. Transistor Sizing for Different k_n/k_p Ratios

The transconductance ratio of the CMOS inverter is given by the formula:

$$\frac{k_n}{k_p} = \frac{k'_n \cdot W_n \cdot L_p}{k'_p \cdot W_p \cdot L_n} \tag{1}$$

Given the technology parameters for the TSMC 0.25 μm process:

- $L_n = L_p = 0.25\,\mu\text{m}$
- $k'_n = 2.501 \cdot 10^{-4}\,\text{A/V}^2$
- $k'_p = 5.194 \cdot 10^{-5}\,\text{A/V}^2$

By keeping $L_n = L_p$, the ratio simplifies to:

$$\frac{k_n}{k_p} = \frac{k'_n \cdot W_n}{k'_p \cdot W_p} \implies \frac{W_n}{W_p} = \frac{k_n}{k_p} \cdot \frac{k'_p}{k'_n} \tag{2}$$

We decide to use the value $W_p = 10\,\mu$m,so the required W_n values are calculated as follows:

1. **Case A:** $k_n/k_p = 1$

$$\frac{W_n}{W_p} = 1 \cdot \frac{5.194 \cdot 10^{-5}}{2.501 \cdot 10^{-4}} \approx 0.207 \implies W_n = 2.07\,\mu\text{m}$$

2. **Case B:** $k_n/k_p = 0.25$

$$\frac{W_n}{W_p} = 0.25 \cdot 0.207 = 0.05175 \implies W_n = 0.5175\,\mu\text{m}$$

3. **Case C:** $k_n/k_p = 4$

$$\frac{W_n}{W_p} = 4 \cdot 0.207 = 0.828 \implies W_n = 8.28\,\mu\text{m}$$

2. Simulation Results and VTC Analysis

After performing the DC sweep simulations in LTspice for the three different k_n/k_p ratios, we obtained the Voltage Transfer Characteristic (VTC) curves (figure 4). From these plots, we determined the critical voltage points: the logic input low voltage (V_{IL}), the logic input high voltage (V_{IH}), and the switching threshold voltage (V_M).

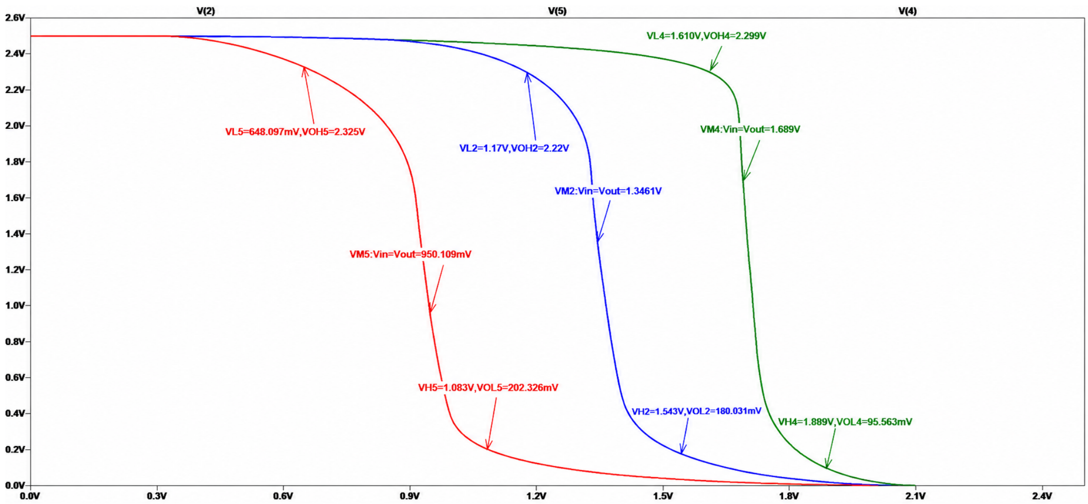


Figure 1: Voltage Transfer Characteristics (VTC) of the CMOS inverter for different k_n/k_p ratios (1, 0.25, and 4).

The switching threshold V_M is the point where $V_{in} = V_{out}$. The points V_{IL} and V_{IH} are defined where the slope of the VTC curve is equal to -1 ($\frac{dV_{out}}{dV_{in}} = -1$).

The extracted values from the simulation are summarized in the following table:

Table 1: Critical DC parameters for different transconductance ratios.

Case	k_n/k_p	V_{IL} (V)	V_{IH} (V)	V_M (V)
A	1	1.17	1.543	1.346
B	0.25	1.61	1.889	1.689
C	4	0.648	1.083	0.950

Based on the Voltage Transfer Characteristics (VTC), we can observe the following:

- When $k_n = k_p$ (Case A), the switching threshold V_M is close to $V_{DD}/2 = 1.25\,\text{V}$, resulting in a relatively symmetric VTC.
- When $k_n < k_p$ (Case B), the VTC shifts to the **right**. Consequently, V_M, V_{IL}, and V_{IH} all increase.
- When $k_n > k_p$ (Case C), the VTC shifts to the **left**. In this case, the transition to a logic low occurs at a lower input voltage, decreasing the values of V_M, V_{IL}, and V_{IH}.

3. Determination of k_n/k_p for 45% Noise Tolerance at Logic 0

The objective is to determine the k_n/k_p ratio that allows the inverter to correctly interpret a logic "0" even when it is corrupted by noise equal to 45% of V_{DD}. This means that the input voltage should be correctly translated as logic "0" until it reaches 45% of the supply voltage (V_{DD}).

Therefore, the input low voltage limit (V_{IL}) must be:

$$V_{IL} = 0.45 \cdot V_{DD} = 0.45 \cdot 2.5 = 1.125\,\text{V} \tag{3}$$

Utilizing LTspice and keeping the width of the PMOS transistor at $W_p = 10\,\mu\text{m}$, we performed successive simulations to adjust the value of W_n. The goal was to find the W_n value for which the derivative of the Voltage Transfer Characteristic ($\frac{dV_{out}}{dV_{in}}$) equals -1 exactly at $V_{in} = 1.125\,\mathrm{V}$ (2)

It was observed that this condition is satisfied when:

$$W_n = 2.38\,\mu\mathrm{m} \tag{4}$$

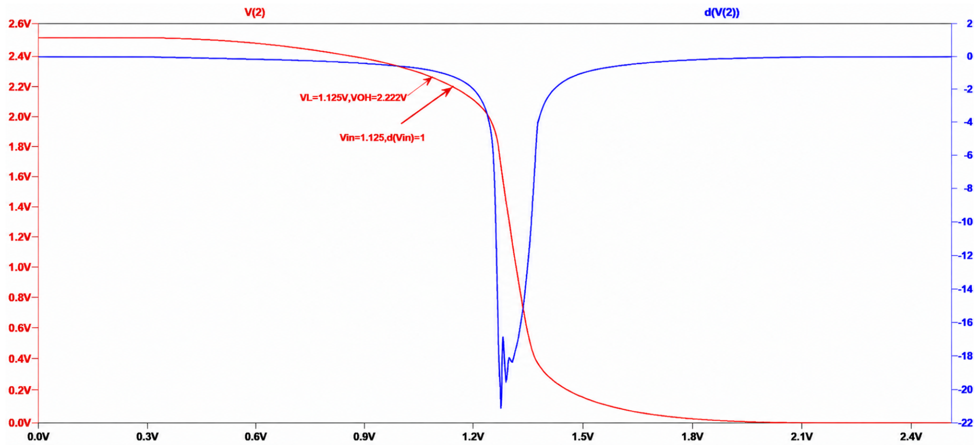


Figure 2: Voltage Transfer Characteristic (VTC) and its derivative (dV_{out}/dV_{in}) for the determination of the V_{IL} point. The cursor marks the point where the slope equals -1 at $V_{in} = 1.125\,\mathrm{V}$, achieved with $W_n = 2.38\,\mu\mathrm{m}$.

The corresponding transconductance ratio k_n/k_p for this sizing is:

$$\frac{k_n}{k_p} = \frac{k'_n \cdot W_n}{k'_p \cdot W_p} = \frac{2.501 \cdot 10^{-4} \cdot 2.38}{5.194 \cdot 10^{-5} \cdot 10} \approx 1.146 \tag{5}$$

Exercise 2

With the help of LTspice, we simulate the behavior of the Voltage Transfer Characteristic (VTC) for various values of V_{DD}. This is achieved by performing successive DC sweep simulations, reducing the supply voltage in steps of $0.05\,\mathrm{V}$ for each simulation, starting from $V_{DD} = 2.5\,\mathrm{V}$ and reaching down to $0\,\mathrm{V}$.

The goal of this analysis is to observe how the reduction of the supply voltage affects the switching threshold (V_M) and the overall gain of the inverter, as well as to determine the minimum V_{DD} for which the circuit still operates as a functional inverter.

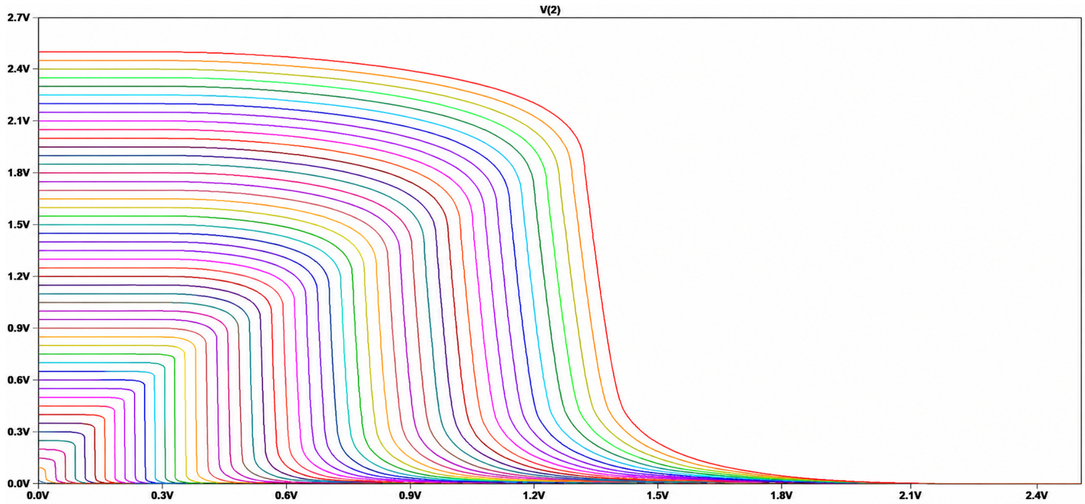


Figure 3: Voltage Transfer Characteristics (VTC) of the CMOS inverter for varying supply voltages (V_{DD}), ranging from $2.5\,\mathrm{V}$ down to $0\,\mathrm{V}$ with a step of $0.05\,\mathrm{V}$. The plot illustrates the compression of the logic levels and the reduction of the transition region slope as V_{DD} decreases.

According to Figure 3, it is evident that as the supply voltage (V_{DD}) drops, the logic output levels (V_{OH} and V_{OL}) decrease as well, due to their direct correlation with V_{DD}. It is observed that the reduction in V_{OH} values is significantly more pronounced, as V_{OH} is directly tied to the decreasing V_{DD}, whereas V_{OL} remains relatively stable and close to $0\,\mathrm{V}$.

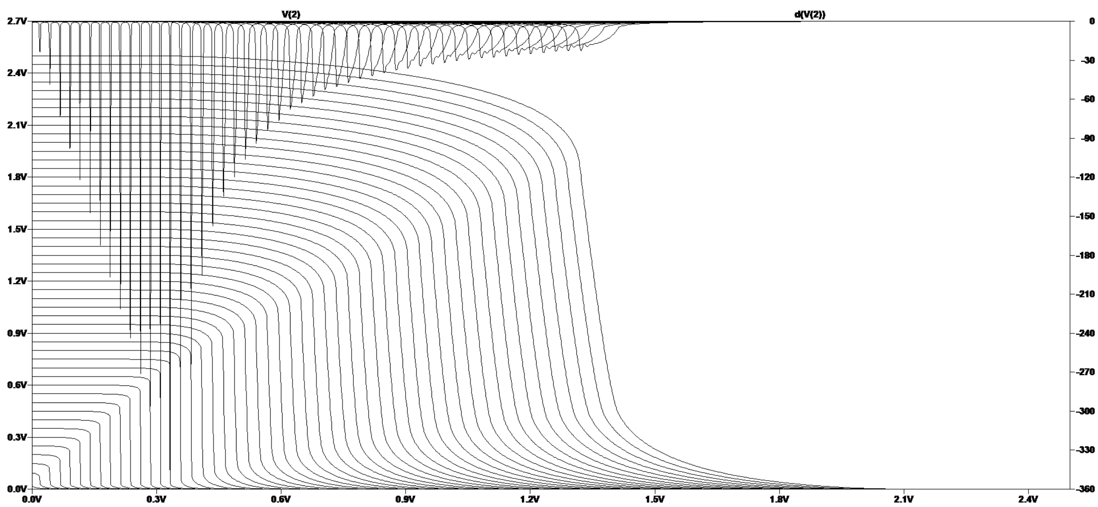


Figure 4: Derivative Graphs for Every Voltage Characteristic

Furthermore, it is observed that the slope of the transition region becomes steeper as the supply voltage (V_{DD}) decreases, with the characteristic curve tending toward a vertical line. This phenomenon is even more evident when examining the derivative graphs ($\frac{dV_{out}}{dV_{in}}$) for various supply voltages in LTspice. Specifically, as V_{DD} decreases, the magnitude of the derivative at the switching threshold (V_M) increases. An exception to this trend occurs when the supply voltage becomes extremely low; at these levels, the inverter's behavior becomes irregular and it eventually ceases to function properly. For instance, for $V_{DD} < 70\,\text{mV}$, the circuit loses its functionality as an inverter, with the output failing to represent valid logic states.

Exercise 3

Using LTspice, we experimented with different ratios between the widths of the nMOS and pMOS transistors to achieve a symmetric Voltage Transfer Characteristic (VTC). Our goal was to secure the switching threshold point (V_M) precisely at the midpoint of the supply voltage ($V_{DD}/2 = 1.25\,\text{V}$).

The widths required to satisfy this condition were determined to be $W_p = 10\,\mu\text{m}$ and $W_n = 2.9\,\mu\text{m}$. This indicates that the width of the pMOS transistor must be approximately 3.4 times larger than that of its nMOS counterpart to compensate for the lower mobility of holes.

The corresponding transconductance ratio is calculated as follows:

$$\frac{k_n}{k_p} = \frac{k_n' \cdot (\frac{W_n}{L_n})}{k_p' \cdot (\frac{W_p}{L_p})} = \frac{250.1 \cdot \frac{2.9}{0.25}}{51.94 \cdot \frac{10}{0.25}} \approx \frac{1450.58}{1038.8} \approx 1.4 \tag{6}$$

This ratio ensures that the current driving capabilities of both transistors are balanced, resulting in a symmetric inverter response (figure 5).

Figure 5 also illustrates that for a symmetrical inverter, the characteristic voltage levels are determined to be $V_{IL} = 1.049\,\text{V}$ and $V_{IH} = 1.432\,\text{V}$.

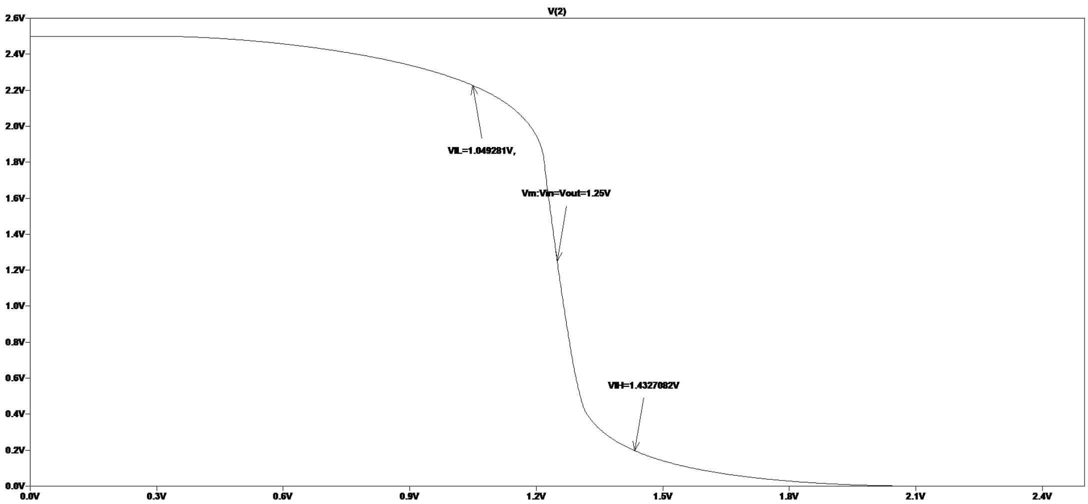


Figure 5: Voltage Transfer Characteristc of a Symmetric Inverter

The next stage of the exercise focuses on the behavioral analysis of the implementation shown in Figure 6. In this configuration, we examine a series of cascaded CMOS inverters to understand how signal characteristics, such as voltage levels and current consumption, evolve as the signal propagates through multiple stages.

Through this setup, we can analyze the regenerative properties of the CMOS logic and how the $V_{in} - V_{out}$ relationship is affected by the cumulative gain of the series connection.

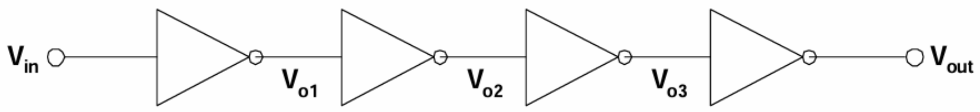


Figure 6: series of cascaded CMOS inverters

In the first case, the input voltage V_{in} is set to the value of V_{IH}. Since the implementation consists of four identical symmetrical inverters, the initial input voltage is defined as $V_{in} = V_{IH} = 1.432\,\text{V}$ as we concluded earlier in this exercise.

Using LTspice, we obtained the following simulation data (Figure 7). The input and intermediate node voltages are defined as follows: The initial input voltage is $V(1) = V_{in} = 1.432\,\text{V}$. The subsequent voltages represent the outputs of each cascaded stage, with $V(2)$ being the output of the first inverter (V_{o1}), $V(4)$ the second (V_{o2}), $V(5)$ the third (V_{o3}), and $V(6)$ the final output (V_{out}). Additionally, $V(3)$ represents the constant supply voltage $V_{DD} = 2.5\,\text{V}$.

The results are summarized in the table below:

Table 2: Voltage levels at each stage of the cascaded inverter chain for $V_{in} = V_{IH}$.

Node	Symbol	Voltage Value (V)
$V(1)$	V_{in}	1.432
$V(2)$	V_{o1}	0.197
$V(4)$	V_{o2}	2.49
$V(5)$	V_{o3}	$8.92 \times 10^{-9} \approx 0$
$V(6)$	V_{out}	2.5
$V(3)$	V_{DD}	2.5

Based on the simulated data, we observe a significant signal restoration across the cascaded stages.Being more precise at the second stage, the output voltage (V_{o2}) reaches $2.49\,\text{V}$, nearly equal to V_{DD}. This indicates that a strong logic "1" has been restored for the input of the third inverter.Additionally the output of the third stage (V_{o3}) is approximately $8.92\,\text{nV}$, which effectively constitutes a strong logic "0".Lastly, the fourth stage output (V_{out}) is fully restored to $2.5\,\text{V}$, equal to the supply voltage.

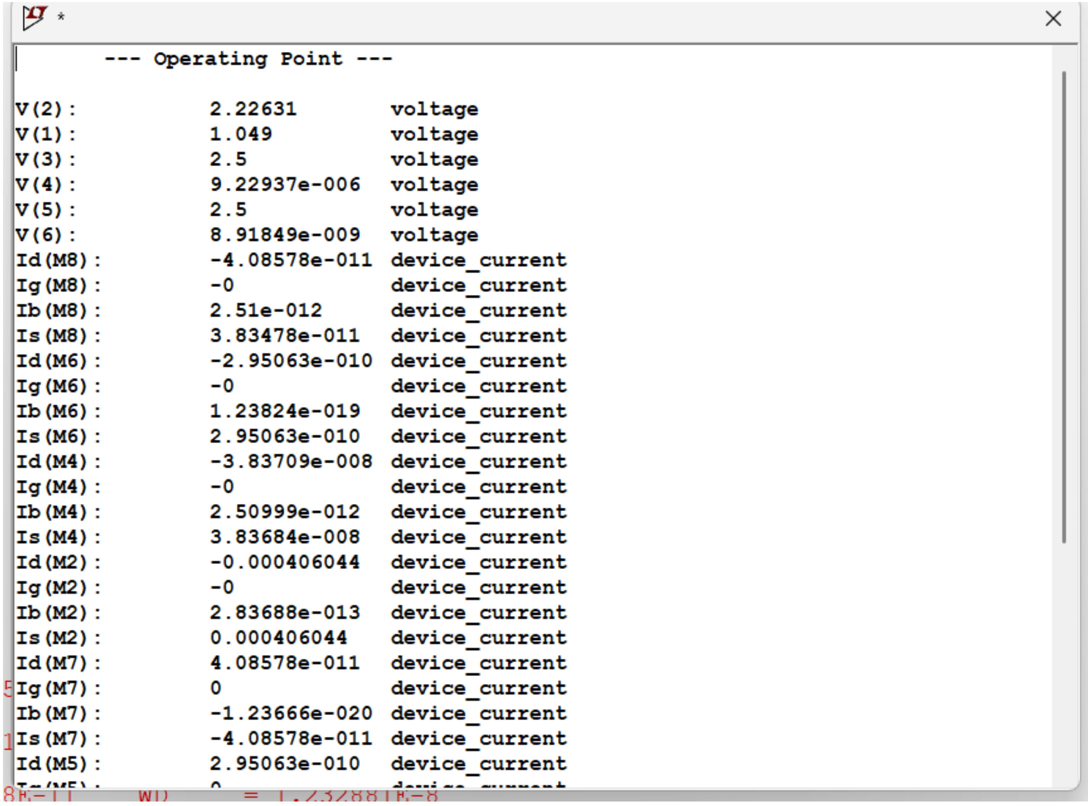
```
--- Operating Point ---

V(2):       0.197305        voltage
V(1):       1.432           voltage
V(3):       2.5             voltage
V(4):       2.49998         voltage
V(5):       8.92247e-009    voltage
V(6):       2.5             voltage
Id(M8):     -2.9499e-010    device_current
Ig(M8):     -0              device_current
Ib(M8):     1.23793e-019    device_current
Is(M8):     2.9499e-010     device_current
Id(M6):     -4.08758e-011   device_current
Ig(M6):     -0              device_current
Ib(M6):     2.51e-012       device_current
Is(M6):     3.83658e-011    device_current
Id(M4):     -6.07123e-008   device_current
Ig(M4):     -0              device_current
Ib(M4):     2.74309e-017    device_current
Is(M4):     6.07123e-008    device_current
Id(M2):     -0.000365781    device_current
Ig(M2):     -0              device_current
Ib(M2):     2.31269e-012    device_current
Is(M2):     0.000365781     device_current
Id(M7):     2.9499e-010     device_current
Ig(M7):     0               device_current
Ib(M7):     -2.51e-012      device_current
Is(M7):     -2.9248e-010    device_current
Id(M5):     4.08758e-011    device_current
```

Figure 7: Operating point voltage data for the cascaded inverter chain, obtained from LTspice Operating Point (.op) analysis for $V_{in} = 1.432\,\text{V}$. The values demonstrate the signal restoration across successive stages.

Following the same logic, we now examine the case where the input voltage is set to V_{IL}. Since we are utilizing the same chain of symmetrical inverters, the initial input value is defined as $V_{in} = V_{IL} = 1.049\,\text{V}$. The subsequent voltages represent the same nodes as in the previous analysis, and their new values are summarized in the table below.The data provided by LTSpice are shown below (figure 8)

Table 3: Voltage levels at each stage of the cascaded inverter chain for $V_{in} = V_{IL}$.

Node	**Symbol**	**Voltage Value (V)**
$V(1)$	V_{in}	1.049
$V(2)$	V_{o1}	2.22
$V(4)$	V_{o2}	$9.22 \times 10^{-6} \approx 0$
$V(5)$	V_{o3}	2.5
$V(6)$	V_{out}	$8.91 \times 10^{-9} \approx 0$
$V(3)$	V_{DD}	2.5



```
--- Operating Point ---

V(2):        2.22631        voltage
V(1):        1.049          voltage
V(3):        2.5            voltage
V(4):        9.22937e-006   voltage
V(5):        2.5            voltage
V(6):        8.91849e-009   voltage
Id(M8):      -4.08578e-011  device_current
Ig(M8):      -0             device_current
Ib(M8):      2.51e-012      device_current
Is(M8):      3.83478e-011   device_current
Id(M6):      -2.95063e-010  device_current
Ig(M6):      -0             device_current
Ib(M6):      1.23824e-019   device_current
Is(M6):      2.95063e-010   device_current
Id(M4):      -3.83709e-008  device_current
Ig(M4):      -0             device_current
Ib(M4):      2.50999e-012   device_current
Is(M4):      3.83684e-008   device_current
Id(M2):      -0.000406044   device_current
Ig(M2):      -0             device_current
Ib(M2):      2.83688e-013   device_current
Is(M2):      0.000406044    device_current
Id(M7):      4.08578e-011   device_current
Ig(M7):      0              device_current
Ib(M7):      -1.23666e-020  device_current
Is(M7):      -4.08578e-011  device_current
Id(M5):      2.95063e-010   device_current
```

Figure 8: Operating point voltage data for the cascaded inverter chain, obtained from LTspice Operating Point (.op) analysis for $V_{in} = 1.049\,\mathrm{V}$. The values demonstrate the signal restoration across successive stages.

Taking everything into consideration, the sequence of inverters successfully regenerates the signal, converting the initial noisy or intermediate input in both cases (V_{IH}, V_{IL}) into solid logic levels in the final output.